The Cost Disease Costs a Tenth of a Point

Baumol's prediction in US consumer spending, 1960-2025

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Abstract

Health care and education rose from 6 percent to 20 percent of US consumer spending between 1960 and 2025, while their price relative to all consumption rose by a factor of 2.7. Baumol's cost disease predicts this pattern when demand for services with slow productivity growth is price-inelastic. Across 72 categories of consumer spending, a category whose relative price rises by 10 percent over a decade gains about 3 percent in budget share, an elasticity of substitution of 0.72 (standard error 0.13). Goods against services give 0.65 in annual data. The relation holds in share-weighted decade changes but not in the 65-year long difference, over which rising income also moves budget shares. The cost to growth is small. If relative prices track relative productivity, the shift of spending away from 1960 weights lowered measured productivity growth in consumption by 0.11 percentage points a year on average, 7.3 percent cumulatively over 65 years, with most of the drag in 1980-99 and none since 2020.

1 The prediction and the data

Baumol (1967) split the economy into a progressive sector, where productivity grows, and a stagnant sector, such as live performance, teaching or nursing, where it does not. Wages equalize across sectors, so the relative price of stagnant output rises at the rate of the productivity gap. Whether the stagnant sector's share of spending rises is governed by σ, the elasticity of substitution between categories. With CES demand, the change in the log budget share of category i is $(1-\sigma)$ times the change in its log relative price, plus a common term. The cost disease requires $\sigma < 1$, the case Ngai and Pissarides (2007) use to generate structural change toward services. The slope of share changes on relative price changes is therefore a direct estimate of $1-\sigma$.

We use the Bureau of Economic Analysis personal consumption expenditure tables 2.4.4 (price indexes) and 2.4.5 (nominal spending), annual, 1959-2025. We keep the most detailed categories that have both a price and a spending series from 1960, which drops internet access (from 1988). We drop the two foreign-travel adjustment lines, and we enter the spending of nonprofit institutions as their final consumption (line 111 of Table 2.4.5) in place of its two components, gross output and the subtracted receipts from sales. That leaves 72 categories, whose shares we renormalize to sum to one. Relative prices are category price indexes divided by the overall PCE price index. For the goods-services split we use the aggregate series in Tables 2.3.4 and 2.3.5.

Specification	Slope $1-\sigma$	Implied σ	N
72 categories, pooled decade changes, share-weighted, decade effects	0.28 (0.13)	0.72	504
excluding tobacco and fuel oil	0.29 (0.14)	0.71	490
unweighted	0.16 (0.11)	0.84	504
72 categories, 1960-2025 long difference, share-weighted	0.11 (0.13)	0.89	72
excluding tobacco and fuel oil	0.19 (0.15)	0.81	70
unweighted	−0.13 (0.15)	1.13	72
Goods vs services, annual changes in log odds, 1961-2025	0.35 (0.13)	0.65	65

Table 1: Change in log budget share on change in log relative price. Pooled decade regressions cluster standard errors by category; long differences use heteroskedasticity-robust standard errors; the goods-services regression uses Newey-West standard errors with three lags and the log ratio of services to goods spending as the dependent variable. Weights are the initial budget share. Data from BEA NIPA Tables 2.3.4, 2.3.5, 2.4.4 and 2.4.5.

2 Results

Spending shifts toward categories whose relative price rises, as the cost disease predicts (Table 1). In decade changes weighted by initial share, with a separate intercept for each decade, the slope is 0.28 with a standard error of 0.13, so σ is about 0.72. Removing tobacco and fuel oil, whose relative prices rose mainly through excise taxes and oil shocks, leaves the estimate unchanged. Aggregating to goods and services gives a similar number. Services rose from 47 percent to 69 percent of consumer spending as their price relative to goods tripled, and annual changes in the services-goods spending ratio load on annual changes in their relative price with a slope of 0.35 (0.13), so σ is 0.65. Health and education are the clearest case. Their nine categories (hospitals, physician, dental and paramedical services, nursing homes, net health insurance, higher education, nursery to secondary schools, and commercial and vocational schools) rose from 6.3 percent to 20.4 percent of spending, and their share-weighted relative price rose by a factor of 2.7. Electronics and telephone equipment saw the largest relative price falls, and higher education, schools and hospitals among the largest rises.

The evidence is weaker in two places. Without weights the decade slope falls to 0.16 (0.11), so the result comes mostly from the large categories, which matter for the aggregate and whose price and share series are measured best. The 65-year long difference has a smaller, insignificant slope, and a negative one without weights, probably because over long horizons rising income moves shares independently of prices. Between 1960 and 1990 the services share rose by more than the relative price change alone can explain at any σ between zero and one, the income effect that Comin, Lashkari and Mestieri (2021) find in structural change across countries. Decade changes with decade effects remove the common part of that shift, though they do not separate prices from income at the category level.

The growth cost of the shift is small. If relative productivity growth shows up one-for-one in falling relative prices, the productivity growth of consumption measured with current spending weights falls short of growth measured with 1960 weights by $\sum_i (\bar{s}_{it} - s_{i,1960})\, \Delta \ln(p_{it}/P_t)$, where $\bar{s}_{it}$ is the average of category i's budget share in years $t-1$ and t and p_{it}/P_t is its relative price. That gap averages 0.11 percentage points a year over 1961-2025 and adds up to 7.3 percent over the period, and it is the same when P_t is a Törnqvist index of the 72 categories in place of the PCE price index. It was 0.32 points a year in 1980-99 and -0.14 in 2020-25. The share of spending in categories whose relative price rose over the whole period went from 68 percent to 73 percent. Nordhaus (2008) reaches a similar conclusion with industry output data, that the cost disease is present but its effect on aggregate growth is modest.

3 Scope

The drag calculation treats relative prices as a measure of relative productivity, which assumes competitive pricing and common wage growth across categories. Health and education prices also reflect insurance, subsidies and market power, and quality change in these categories is poorly measured, so the rise in their relative price may overstate their productivity gap. The elasticity estimates come from category-level demand and do not identify a supply-side cause of the price changes.

References

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